Sankarshan Kulkarni

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Professional Summary

Quantitative researcher with deep expertise in **options pricing**, **volatility surface modelling**, **systematic backtesting**, and **risk analytics**. Proficient in **Python**, **SQL**, and **statistical methods**. Proven track record of building high-performance algorithmic trading systems and risk frameworks for multi-billion dollar portfolios.

Work Experience

Associate (Y3) – PricewaterhouseCoopers Services LLP, India Mar 2026 – Present

Bengaluru, Karnataka

- Built quantitative risk analytics and forecasting frameworks for **$10B+** AUM hedge fund portfolios across Equities, Fixed Income, MBS, and Alternative Investments.
- Developed **Python/SQL** models for risk forecasting and portfolio analytics across **100K+** instruments, reducing manual effort by **1000+ hours annually**.

Technology Analyst (C10) – Citibank (CSIPL) Jul 2024 – Feb 2026

Pune, Maharashtra

- Designed scalable **Python**, **Spark**, and **SQL** data pipelines for quantitative reporting and predictive analytics on terabyte-scale financial datasets.
- Engineered analytical infrastructure and feature engineering for **100+** datasets, improving data quality and research productivity.

Key Projects

BankNifty Dispersion Trade Monitor GitHub

- Developed a real-time dispersion monitor using **WebSocket streaming** (Zerodha KiteTicker) and **multi-threaded API calls** to track net premium across index and constituent straddles, normalised to a **Rs. 6 crore** reference portfolio.
- Achieved **75% reduction** in API calls via smart caching and implemented automatic fallback mechanisms (WebSocket, polling, API, mock) for uninterrupted data delivery.

Short Strangle Backtesting (BankNifty) GitHub

- Built a high-performance 1-min intraday backtest engine processing **10M+** rows of options data; optimised vectorised logic reduced runtime **less than 45 seconds**.
- Computed **Sharpe ratio (1.3)**, **CAGR (8%)**, **max drawdown**, skewness (-1.36), and kurtosis (0.37); validated risk-adjusted performance across 1+ year of tick data.

SVI Volatility Surface Modelling (SPX) GitHub

- Extracted OTM SPXW options (0/1/2 DTE) from a **7.5M-row** dataset; derived forward price using 10-year Treasury yield (r=4.5%) and computed **Black-76 implied volatilities**.
- Fitted a **5-parameter SVI** model via least-squares optimisation, achieving **RMSE 0.00256**; verified no butterfly arbitrage via call price convexity.

Kalman Filter Pair Trading GitHub

- Developed a statistical arbitrage strategy on US equities using **Kalman Filter state-space modelling** for dynamic hedge ratio estimation; backtested on 5+ years of daily data.
- Achieved **Sharpe 1.20**, **Calmar 2.0**, max drawdown **20%**, and positive net returns after transaction costs and slippage.

Risk Parity & Minimum Volatility Portfolio GitHub

- Implemented two optimisation methods – risk parity (marginal risk contribution) and min-vol with product constraint – on six US stocks using 1-month historical returns.
- Achieved daily variance **10× lower** than market-cap weighted benchmark; risk parity weights evenly distributed (0.12–0.20), confirming true diversification.

Optimised Intraday EMA-RSI Strategy GitHub

- Engineered a **Numba-JIT** accelerated backtesting engine (**5–10x speedup**) with multi-timeframe trend indicators (EMA) and momentum filter (RSI); tuned thresholds via grid search.
- Built an interactive **Streamlit** dashboard providing real-time performance metrics, drawdown analysis, and Sharpe tracking; optimised memory usage for large intraday datasets.

Black-Scholes PDE – Finite Difference GitHub

- Transformed the Black-Scholes PDE to the heat equation and solved it using an **explicit finite difference** scheme on a discretised log-moneyness domain.
- Validated against the analytical Black-Scholes formula, achieving $\mathbf{R^2 = 0.995}$; quantified numerical convergence and stability (CFL condition).

Education

2020 – 2024 B.Tech, Chemical Engineering, **Indian Institute of Technology Gandhinagar**

Technical Skills

Languages & Tools	**Python**, **SQL**, C++, Matlab, Spark, **Pandas**, **NumPy**, **SciPy**, **Statsmodels**, Docker, Git, Linux, Streamlit, Numba, Flask, WebSockets
Quantitative Domains	Factor Research, Portfolio Construction, Risk Modeling, Statistical Arbitrage, Time Series Analysis, Monte Carlo Simulation, Derivatives Pricing (Black-Scholes, Black-76), CAPM, Risk Parity, **SVI Volatility Modelling**, Backtesting, Implied Volatility, Finite Difference Methods

Achievements

- Global Rank **87** among **4000+** participants in NK Securities Quant Research Challenge for volatility surface construction and option mispricing identification.
- Winner – Algorithmic Trading Hackathon among **2000+** participants for designing systematic quantitative trading strategies and robust statistical backtesting frameworks.